MALMÖ UNIVERSITY
FACULTY OF CULTURE
AND SOCIETY

Queer(ing) Interaction Design:

Proposing Orientation As a Design Material And Facilitating Somatic Consciousness

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Abstract

This thesis investigates how Interaction Design can facilitate somatic consciousness of orientation within institutional "straightness" through embodied interaction, ambiguity, and non-normative movement. Drawing on queer phenomenology, somaesthetic interaction design, and Research-through-Design practices, the project explores orientation as a socio-somatic design material shaped through bodily movement, institutional expectations, visibility, and spatial relations. Through participatory workshops, reflective drawings, embodied digital sketching, and responsive sound-based interventions situated within university spaces, participants were encouraged to negotiate movement beyond habitual and efficiency-driven behaviour. The findings suggest that refusing normative movement patterns can provoke hesitation, playful experimentation, and reflection on bodily visibility and socially acceptable behaviour within institutional environments. The thesis offers two connected contributions: first, orientation as a design material for Interaction Design; and second, a social contribution by exploring how design can facilitate somatic consciousness of institutional orientations through embodied and non-normative movement.

Keywords: Queer Phenomenology, Embodied Interaction, Ambiguity, Normativity, Queer HCI, Interaction Design, Design Material

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1. INTRODUCTION: AIM AND RESEARCH QUESTION

1.1 Background

Interaction Design increasingly attends to embodied, affective, and situated forms of interaction rather than only functional or efficiency-driven systems (Dourish, 2001; Höök, 2018). This thesis builds on that shift by asking how interaction can make bodily orientation within institutional environments more noticeable, negotiable, and open to reflection.

The project brings Interaction Design into dialogue with Sara Ahmed's Queer Phenomenology (2006), where orientation is understood as the way bodies are directed through repeated relations with spaces, objects, and social expectations. From this perspective, institutional environments such as universities are not neutral settings. Hallways, classrooms, entrances, and transitional spaces subtly shape how bodies move, where they are expected to go, and what kinds of behaviour feel appropriate.

This thesis investigates these institutional orientations through embodied workshops and sound-based interaction. Rather than designing for smoothness, efficiency, or seamless movement, the project explores how refusal, ambiguity, and non-normative movement can make habitual spatial behaviours more perceptible (Gaver et al., 2003). In doing so, it asks whether orientation can be approached as a socio-somatic design material within Interaction Design and how such a material may contribute to Queer HCI (Spiel et al., 2019; Taylor et al., 2024; Park et al., 2025).

1.2 Motivation

A core motivator for this work is the political and methodological act of queering Human Computer Interaction (HCI) (Park et al., 2025; Spiel et al., 2019; Taylor et al., 2024). Standardised computer systems often enforce normative ways of being, prioritising speed and efficiency. This project is motivated by a deliberate resistance to these mandates, embracing fluidity and queer interaction states (Park et al., 2025; Spiel et al., 2019; Taylor et al., 2024). Moreover, further motivation for this project emerges from an interest in how institutional environments shape bodily behaviour and how interaction design might challenge these normative conditions. Inspired by Ahmed's work on orientation and normativity, this thesis investigates how everyday movements within university spaces become structured by implicit expectations regarding efficiency, direction, and appropriate behaviour. Hallways, classrooms, entrances, and transitional spaces guide bodies along particular paths that gradually become habitual and largely unnoticed (Marshall et al., 2023).

As designers, this raises important questions regarding the role of interaction design in either reinforcing or challenging such structures. Much contemporary digital interaction still prioritises optimisation, usability, and frictionless experience. While these approaches have clear practical value, they may also reproduce normative assumptions regarding how bodies should move, behave, and interact within designed systems.

The project is further motivated by queer theoretical perspectives that view disruption and failure as productive rather than negative conditions. Drawing on Halberstam's *The Queer Art of Failure* (2011), the research approaches disorientation not as a design flaw, but as a potential site for reflection and resistance. Through this lens, moments where systems fail to support normative movement become opportunities for participants to renegotiate their relationship with institutional space and with their own bodily orientation.

To investigate this, the thesis combines theoretical inquiry with practical design exploration through self-directed workshops and embodied interventions situated within university environments. The workshops utilise movement, spatial negotiation, and audio

feedback to examine how participants experience and respond to disruptions of normative orientation. In this way, theory and practice become closely interconnected: queer phenomenology provides the conceptual grounding for understanding orientation and normativity, while interaction design methods inspired by ambiguity and embodied interaction provide practical ways of exploring these concepts experientially.

The central research questions guiding this thesis are:

1. How can interaction design facilitate a somatic consciousness of orientation in institutional spaces by refusing to respond to normative movement?
2. In what ways can orientation be a socio-somatic design material?

Through these questions, the thesis aims to contribute to Interaction Design by proposing orientation as a socio-somatic design material that can be worked with through interactive and embodied practices. Furthermore, the project contributes methodological insights through the mentioned workshops that utilise disorientation, ambiguity, and bodily negotiation as design strategies. Rather than treating interaction as a process of optimisation alone, this research positions interaction design as a critical and reflective practice that questions how bodies are permitted to inhabit institutional space.

1.3 Ethical Considerations and AI Use

This thesis follows the ethical principles of good research practice outlined by the Swedish Research Council (2017), alongside the European Union General Data Protection Regulation (GDPR) (European Union, 2016). Because the project investigates embodied interaction, bodily orientation, and participant experiences within institutional environments, ethical considerations extend beyond data handling alone and include questions of autonomy, integrity, accountability, and respectful treatment of participants throughout the research process.

The workshops involved movement-based interaction (Marshall et al., 2023), embodied sketching (Márquez Segura et al., 2016), and semi-structured interviews (Kvale & Brinkmann, 2015) situated within university spaces. All participants were informed about the purpose and structure of the study prior to participation, including the use of camera-based tracking systems within the interactive installation. Participation remained voluntary throughout the process, and participants were informed of their right to withdraw from the study at any stage without explanation. Any visual documentation included in this thesis was used only with explicit participant consent, and identifying details were anonymised where necessary in order to protect participant privacy.

While the installation utilised camera tracking technologies through MediaPipe and TouchDesigner (Derivative, 2026), the system was designed as a real-time interactive system rather than a surveillance or biometric identification system. No facial recognition, personal identification, or long-term behavioural tracking was conducted. Movement and positional data were processed temporarily, but not recorded or analysed, for interaction purposes and were not stored in ways that could identify individual participants.

Artificial intelligence (AI) tools were used selectively throughout the research and writing process as supportive tools rather than autonomous research agents. AI-assisted tools were primarily used for interview transcription, structuring and organising reflective notes, and receiving technical assistance during the TouchDesigner sketching process. However, all analytical interpretations, theoretical overview, workshop design decisions, and written arguments remained critically evaluated and written by the student.

Particular attention was given to the limitations and biases inherent within AI systems and training datasets. AI-generated outputs were not accepted uncritically and were continuously reviewed, revised, and contextualised in relation to the project's theoretical and methodological framework. The thesis, therefore, approaches AI not as a neutral or objective system, but as a technological tool requiring ongoing critical reflection, accountability, and researcher responsibility.

2. THEORIES AND PRACTICES

2.1 Shifts in Interaction Design

This thesis contributes to strands within Interaction Design and Human–Computer Interaction that approach interaction as embodied, situated, and shaped through lived experience rather than optimisation alone. Building on embodied interaction (Dourish, 2001), somaesthetic interaction design (Höök, 2018; Höök & Löwgren, 2012), and Queer HCI (Park et al., 2025; Spiel et al., 2019; Taylor et al., 2024), the project understands interaction as a means of offering deeper somatic consciousness, reflection, and alternative ways of inhabiting institutional space, rather than prioritising seamlessness and efficiency.

Researchers such as Dourish (2001) argued for embodied interaction, emphasising that interaction does not occur separately from the body and the world, but emerges through situated bodily engagement. This shift repositioned experience, movement, and material relations as central concerns for design practice.

At the same time, Soma Design introduced approaches that treat the living, sensing body as a site of knowledge and design practice (Höök & Löwgren, 2012; Höök, 2018). Rather than reducing bodily experience to measurable data, somaesthetic approaches focus on felt sensation, movement, and affect. Höök describes bodily movement and physiological signals as malleable design materials that can be shaped, explored, and reflected upon through interaction. In this context, interaction becomes not only a functional exchange between human and system, but also an affective and experiential relation capable of reshaping somatic consciousness.

Parallel to this movement, critical and reflective approaches within Interaction Design challenged the dominance of clarity and usability. Gaver et al. (2003) proposed ambiguity as a resource for design, arguing that uncertainty and incomplete interpretation can encourage reflection, participation, and meaning-making. Rather than eliminating friction, these approaches intentionally utilise it to destabilise habitual interaction patterns.

This movement resonates with queer theoretical perspectives that later informed Queer HCI, particularly Halberstam's (2011) concept of failure as a productive and disruptive force. Drawing from queer theory and critical theory, *The Queer Art of Failure* reframes friction, disorientation, and breakdown not as conditions to overcome, but as opportunities to resist normative expectations and open alternative forms of experience and participation.

The present thesis situates itself within these shifts in Interaction Design by investigating how bodily orientation, uncertainty, and non-normative movement may function as materials for interactional exploration and further inform Queer HCI. Inspired by Zimmerman and Forlizzi's (2014) model of Research through Design in HCI, this diagram (Figure 1) positions Interaction Design as a mediating practice between theoretical foundations, Queer HCI, and the thesis contribution of orientation as a possible design material.

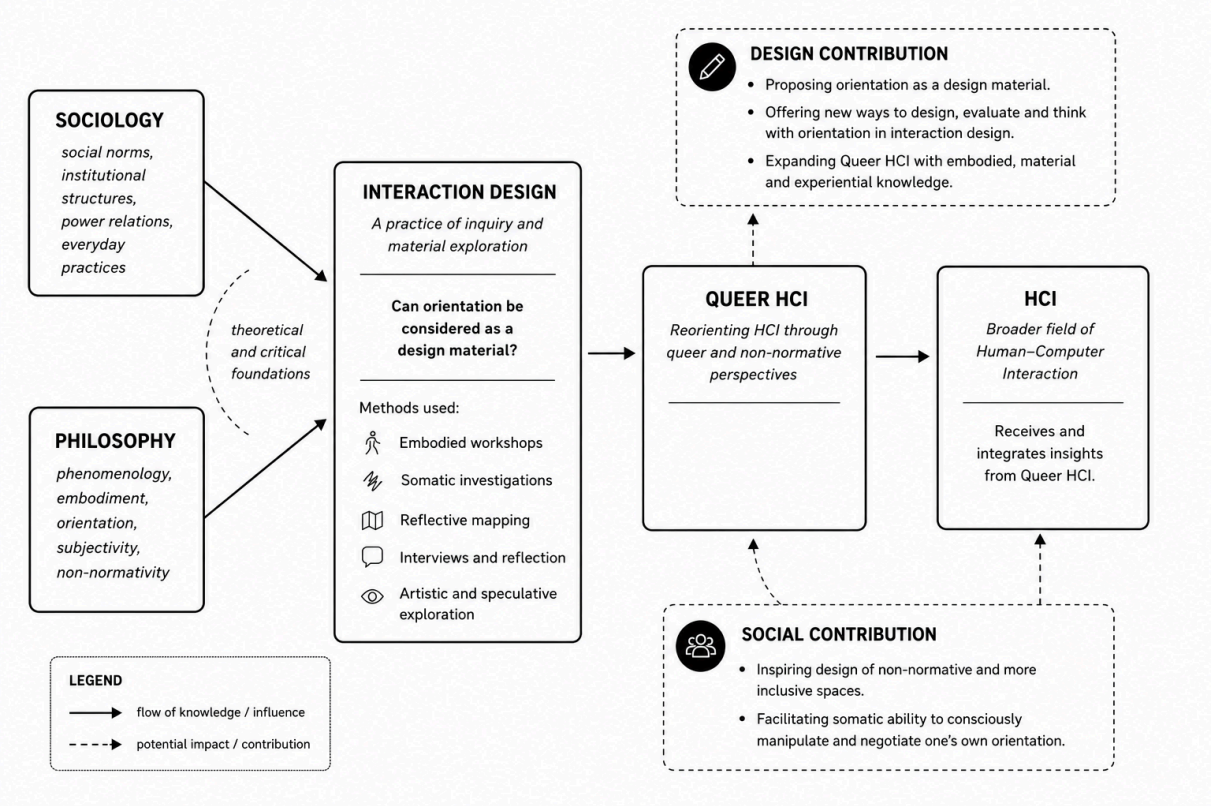


Figure 1 — *Visual representation of the disciplinary connections*

2.2 Queer Phenomenology, Orientation, and Non-Normativity

To investigate bodily orientation within institutional environments, this thesis draws primarily on Sara Ahmed's *Queer Phenomenology* (2006). Ahmed expands phenomenological understandings of orientation by examining how bodies are directed through repeated interactions with spaces, objects, and social structures, which has a growing presence in interaction design. Orientation, in this sense, is not merely spatial positioning, but a relational process through which certain paths, actions, and behaviours become available, familiar, and socially reinforced.

Maurice Merleau-Ponty's phenomenology also becomes important within Ahmed's discussion of orientation. In *Phenomenology of Perception* (2002), Merleau-Ponty argues that bodily experience is always relational and situated within the world rather than detached from it. Perception emerges through embodied engagement with surrounding environments, objects, and spatial relations. Ahmed builds upon these phenomenological ideas while simultaneously queering them, shifting attention toward how bodies are socially and institutionally directed along normative lines. In this sense, orientation is not only about how bodies inhabit space phenomenologically, but also about how social norms shape which directions, movements, and ways of inhabiting space become available, naturalised, or restricted.

Ahmed argues that institutions organise bodies along normative lines. These lines shape what is considered natural, appropriate, and efficient movement, while simultaneously marginalising movements and orientations that deviate from such expectations. Bodies become oriented through repetition; over time, institutional spaces subtly guide individuals toward particular behaviours and ways of inhabiting space.

Within university environments, these orientations emerge through architecture, circulation paths, classroom structures, schedules, and social expectations. Hallways,

entrances, desks, and transitional spaces are not neutral containers for movement, but active participants in shaping how bodies navigate institutional life. Ahmed describes this process as "straightening," where bodies are repeatedly directed toward socially accepted lines and modes of behaviour.

Queer phenomenology becomes particularly relevant here because it attends to moments where bodies fail to align with these normative directions. Rather than viewing deviation as error, Ahmed frames disorientation as a productive condition capable of revealing otherwise invisible structures of normativity. Non-normative movement exposes the lines that normally remain unnoticed precisely because they are habitual and socially reinforced.

This perspective aligns closely with the interests of the present thesis. The project investigates how interaction design may create situations where participants use their somatic orientation consciously through moments of disruption, friction, and bodily negotiation. By interrupting habitual movement patterns, interaction can potentially reveal how bodies are shaped by the institutional environments they inhabit.

2.3 Shifting Toward Interaction Design Materials and Methods

The theoretical perspectives discussed above do not remain solely conceptual within this thesis; they directly inform the practical and methodological direction of the project. The shift toward embodiment, ambiguity, and non-normativity within Interaction Design opens possibilities for approaching design not as the production of fixed solutions, but as an exploratory and material practice.

Within Interaction Design research, material-oriented perspectives have increasingly examined how interaction emerges through relations between bodies, technologies, spaces, and physical materials (Vallgårda & Redström, 2007; Vallgårda, 2009; Wiberg, 2018). Computational systems are not understood as immaterial processes detached from lived experience, but as material composites that gain meaning through situated interaction.

Vallgårda (2009) describes computational composites as relational assemblages formed between digital technologies, physical environments, and human activity. This perspective becomes particularly relevant for the present thesis, where bodily orientation is explored through the intersection of spatial movement, institutional design, sensor-based interaction, and somatic experience. Orientation is therefore approached not simply as a spatial coordinate, but as a socio-somatic condition emerging between body, environment, and system.

Similarly, Löwgren (2016) emphasises the significance of making within Interaction Design research, arguing that design practice itself can function as a mode of inquiry and knowledge production. Designing becomes a way of exploring experiential qualities and relations that may not be fully accessible through theoretical analysis alone. This approach supports the thesis's movement toward embodied workshops and interactive interventions as forms of investigation rather than merely demonstrations of pre-existing theory.

Recent somaesthetic and embodied interaction research has further expanded how bodily experience may function within interaction design practice. Núñez-Pacheco and Rosén (2024) discuss the "felt sense" as a mode of noticing grounded in tacit bodily knowledge, arguing that somatic consciousness can function as a way of attending to subtle relational and affective experiences that often remain difficult to articulate. Rather than privileging optimisation or measurable interaction alone, their work foregrounds slowness, somatic attention, and embodied reflection as important components of design inquiry.

This perspective becomes particularly relevant for the present thesis, where orientation is not approached solely as directional movement through space, but as a bodily and affective condition shaped through institutional environments, visibility, and social expectation. Similar

to Núñez-Pacheco and Rosén's discussion of felt sensing and bodily noticing, this thesis investigates how interaction design may create situations where participants become more attentive, finally asking whether orientation can be more than a design approach, but can be a design material.

2.4 Designing with and for Uncertainty

Building upon Gaver et al. (2003), this thesis approaches uncertainty and ambiguity as productive interactional qualities. Rather than aiming for complete clarity or seamless responsiveness, the project investigates how friction and disorientation may provoke somatic reflection.

In this context, uncertainty is not treated as a technical limitation but as an intentional design strategy. Interactions that refuse immediate legibility or expected feedback can interrupt habitual movement patterns and encourage participants to renegotiate their relation to institutional spaces. Designing with uncertainty, therefore, becomes a way of opening interaction toward reflection, negotiation, and alternative bodily orientations.

Halberstam's (2011) discussion of failure further supports this position by framing deviation and breakdown as productive rather than negative conditions. Failure interrupts expectations of smoothness and progression, opening space for alternative relations, movements, and experiences. In this thesis, non-normative orientation is therefore approached not as dysfunction, but as a critical and experiential possibility within interaction design practice.

This perspective also resonates with somaesthetic approaches, where bodily noticing and affective awareness emerge through subtle shifts in movement, sensation, and attention (Höök, 2018). Moments of hesitation, imbalance, or interruption become valuable experiential conditions rather than interactional failures to eliminate.

2.5 Designing for and with Orientations

The project further proposes orientation itself as a material for interaction design practice. Following Vallgårda's material perspectives and Ahmed's understanding of orientation, bodily direction and spatial relation are approached as designable conditions that can be manipulated, disrupted, and explored through interaction.

The practical exploration of these ideas gradually moves the thesis toward Research-through-Design approaches, where making, experimentation, and embodied participation become central modes of inquiry. The following chapter, therefore, discusses the methodological practices used to investigate orientation through interactive workshops, somatic experimentation, and situated design interventions.

Glöss (2024) identifies a broader challenge within HCI: the disconnect between theoretical aspirations and their translation into design execution. This tension is relevant to this thesis, which draws on queer phenomenology while also needing to develop an interaction design outcome.

This thesis does not claim to resolve this gap. Instead, the workshops are positioned as an attempt to work across it. Concepts such as orientation, straightness, institutional normativity, and somatic consciousness are translated into situated activities: walking, listening, embodied sketching, and reflecting. In doing so, the workshops make theoretical concepts experiential and materially negotiable.

The workshops operate as a practical bridge between queer phenomenology and interaction design practice. They allow orientation to be felt, tested, documented, and discussed as a possible design material, rather than remaining only a theoretical concept (Glöss, 2024; Zimmerman & Forlizzi, 2014).

3. METHODS OVERVIEW

This chapter presents the methodological approaches used throughout the thesis and outlines how the practical investigation was organised. The methods were selected to investigate how interaction design can facilitate somatic consciousness of institutional orientations through embodied and non-normative movement within university environments.

Because the research investigates orientation through lived bodily experience, the methods emphasise participation, movement, interaction, and embodied exploration within institutional environments. Workshops, digitally responsive interventions, movement-based installations, and reflective drawing activities function as the primary methods of inquiry throughout the thesis.

The methods presented in this chapter are followed by the practical implementation of the workshops and interventions conducted at Malmö University.

3.1 Design Ethnography

This thesis adopts a design ethnographic approach informed by situated observation, bodily engagement, and spatial exploration. Design ethnography allows designers to investigate how people interact with environments, objects, and systems while remaining attentive to social, spatial, and affective relations (Pink et al., 2022). Within this project, the university environment is approached not as a neutral setting, but as an active participant in shaping bodily orientation and movement.

The ethnographic approach used in this thesis focuses particularly on observing participants and participants experiencing movement within institutional environments. Attention is placed on bodily rhythms, hesitation, slowing down, directional movement, and moments where normative pathways become disrupted or renegotiated. The project combines observation with embodied participation, allowing the researcher to investigate orientation not only visually, but also through lived bodily experience.

3.2 Phenomenological Investigations

Phenomenological approaches within Interaction Design emphasise lived experience, bodily perception, and situated engagement with the world (Kozel, 2007). In this thesis, phenomenological reflection was implemented through both researcher and participant-led processes. My own reflections were documented through field notes, sketches, diagrams, photographs, and observations made during the workshops. Participants also engaged in reflection through drawing: after moving through the installation, they drew their bodily orientation and movement trajectories on paper. This process follows Kozel's (2015) understanding of process phenomenologies, where embodied experience can be reflected upon through writing, drawing, notation, and other forms of documentation that remain close to lived bodily experience.

The focus of these investigations is not objective measurement, but attentiveness toward how institutional orientations are bodily experienced. Through movement, sensory attention, reflective notation, and embodied observation, the project investigates how participants experience disruption, uncertainty, and non-normative orientation within university space.

Together, these reflective practices align with Hansen and Kozel's (2007) understanding of embodied imagination as a hybrid design method, where movement, bodily experience, and imagination contribute directly to the development of interactive design work.

This approach aligns closely with somaesthetic perspectives within Interaction Design, where bodies and felt sensation become important forms of experiential knowledge (Höök, 2018). Attention is therefore given to bodily hesitation, shifting rhythms, directional awareness, and affective responses emerging during interaction.

3.3 Embodied Digital Sketching

Alongside analogue sketching methods, more importantly, the project incorporates embodied digital sketching as a way of exploring orientation, movement, and interaction through computational systems. Inspired by embodied sketching approaches (Márquez Segura et al., 2016), the process focuses on generating ideas through movement, bodily experimentation, and immediate interaction with digital materials.

TouchDesigner is used as a sketching environment for exploring how bodily orientation, movement speed, gaze direction, and spatial positioning can produce different interactional conditions. Rather than functioning as a finished system, these sketches operate as exploratory and ephemeral experiments.

Embodied digital sketching allows interaction to be explored directly through movement and situated participation rather than solely through conceptual planning. This creates a process where bodily experience, interaction logic, and spatial negotiation develop simultaneously through iterative experimentation.

3.4 Material Exploration

Material exploration within this thesis focuses on orientation, movement, and interaction as experiential and relational conditions rather than fixed technical outputs. Following material-oriented perspectives within Interaction Design (Wiberg, 2013; Vallgårda, 2009), interaction is approached as something emerging between bodies, technologies, spatial conditions, and institutional structures.

This process of material exploration gradually informed the structure and implementation of the participatory workshops conducted within Malmö University.

4. DESIGN PROCESS

This chapter presents the development of the project through a series of theoretical explorations, conversations, embodied experiments, and workshop iterations. Rather than emerging through a linear process, the project evolved gradually through reading, discussion, sketching, testing, and reflective redesign.

The design process moved between theory and practice continuously, where conceptual insights informed practical interventions, while embodied experiments and participant responses reshaped the direction of the research. The process, therefore, became an iterative investigation into how orientation, movement, and institutional norms may be explored through interaction design.

4.1 Literature Review — Orientation and Queering HCI

The design process began through extensive reading and theoretical investigation around orientation, embodiment, queer phenomenology, and Interaction Design. This initial stage functioned as the conceptual origin of the project and established the theoretical grounding that informed the later practical explorations. The results can be found in Chapter 2 above.

4.2 Conversations with Peers

Intention and Premises

Following the literature review, I conducted informal sit-down conversations with peers as an early exploratory step in the design process. The intention was to connect theoretical ideas from queer phenomenology, particularly orientation and institutional straightness, with lived experiences of moving through university spaces. These conversations were not structured as formal interviews, but as open qualitative exchanges that helped me understand how people perceived their bodies, movement, and visibility within institutional environments.

This approach was informed by Swain and King's discussion of informal conversations in qualitative research, where such conversations are described as a useful method for generating more naturalistic and less formal data, particularly when used to complement other qualitative methods (Swain & King, 2022).

Materials

- Notebook
- Researcher notes
- Open conversational prompts
- Literature review notes on orientation and institutional straightness

Number of Participants

Around 10 peers (approximately 8 of the 10 participants self-identified as queer).

Location

Malmö University and informal university-related settings.

Length

Approximately 15 minutes per conversation.

Description

The conversations were guided by open questions around orientation, somatic ability, institutional straightness, and movement within university spaces. Participants were asked how they understood orientation, how aware they were of their bodies in institutional environments, and how they perceived their own movement through university hallways and shared spaces.

A recurring point of discussion was how participants imagined they would move if they were not being observed. This helped open up reflections on visibility, attention, social expectation, and the ways bodies adjust themselves in institutional environments.

Takeaways

The conversations revealed differences in how participants described orientation and spatial perception. Queer participants often described being highly aware of their bodies, the space they occupied, and the attention they received while moving through institutional environments. Several discussed consciously negotiating how they moved, positioned themselves, or adjusted their bodily behaviour depending on the surrounding environment and the perceived gaze of others.

In contrast, many non-queer participants described movement through institutional spaces as more habitual, automatic, functional, or efficiency-oriented. However, across both groups, movement within the university was still described as shaped by visibility, attention, and social expectation. Many participants described trying to avoid attracting attention to themselves, often defaulting toward efficient or "straight" movement patterns.

These conversations became important because they helped connect the theoretical ideas emerging from the literature review with lived accounts of bodily negotiation in institutional space. They supported Ahmed's understanding of orientation as socially and spatially produced, while also suggesting that orientation is experienced not only spatially, but affectively, through visibility, judgment, observation, and self-consciousness.

4.3 Preliminary Workshop — Wizard of Oz Installation

Intention and Premises

The preliminary workshop was conducted as an early embodied exploratory probe (Márquez Segura et al., 2016) into how participants orient themselves within institutional space. The aim was to test whether habitual movement would initially follow the straight, efficient path of least resistance, and whether an interactive intervention could disrupt this pattern. This phase was informed by Ahmed's (2006) concept of institutional straightness, where bodies are directed through repeated and socially reinforced orientations.

The workshop also tested an early Wizard of Oz version of the installation, where the interaction was manually controlled by the author rather than technically automated. This allowed the core interaction logic to be explored before developing the system further in TouchDesigner and MediaPipe.

Materials

- Projected flower video
- Whiteboard projection surface
- Laptop/projector
- Notebook/documentation materials
- Camera/phone for documentation

Number of Participants

3 participants (Participants experienced the installation individually.)

Location

Classroom, Malmö University

Length

Approximately 10 minutes per participant.

Description

Participants began from the opposite side of the classroom, facing the whiteboard. They were first asked to walk toward the board as they normally would. All participants initially walked in a straight line, following the most direct and efficient route as expected.

In the second stage, a projected video of a blooming flower was displayed on the whiteboard. Participants were asked to activate the installation through movement, but were not told how the system worked. The installation did not respond to straight movement. When participants began experimenting with non-normative movement patterns, such as zig-zagging, turning, hopping, or pirouettes, the researcher manually activated the projection response (Figure 2).

After the interaction, short semi-structured interviews were conducted. These focused on somatic abilities, movement, institutional expectations, visibility, and discomfort (Appendix 1). The questions intentionally shifted away from usability and toward somatic experience, asking participants to reflect on how their bodies prepared to move, what they felt when the installation did not respond, and what changed when they began moving differently.

Takeaways

The workshop confirmed that straight movement initially emerged as automatic and unquestioned. Participants described their first instinct as moving from "point A to point B" as efficiently as possible. This supported the premise that institutional spaces often encourage direct, functional, and habitual movement.

The workshop also revealed differences in how participants related to visibility and somatics. Some participants only became aware of institutional expectations after the installation failed to respond and forced them to reconsider their movement. In contrast, the queer participant already showed heightened awareness of visibility, spatial positioning, and the attention their body might receive within the classroom. This suggested that orientation is not only spatial, but also social, affective, and relational. The participants self-identified their sexual orientation after learning more about the thesis project.

A key insight concerned the projected flower video. Although it was initially chosen as an atmospheric visual element, it directly influenced Participant 2, who began performing pirouettes inspired by the blooming motion. This showed that visual feedback was not neutral, but could guide or prescribe participant behaviour. As a result, later iterations moved away from visually suggestive projection and toward sound-based interaction, in order to reduce visual prompting and allow participants to negotiate orientation more openly.

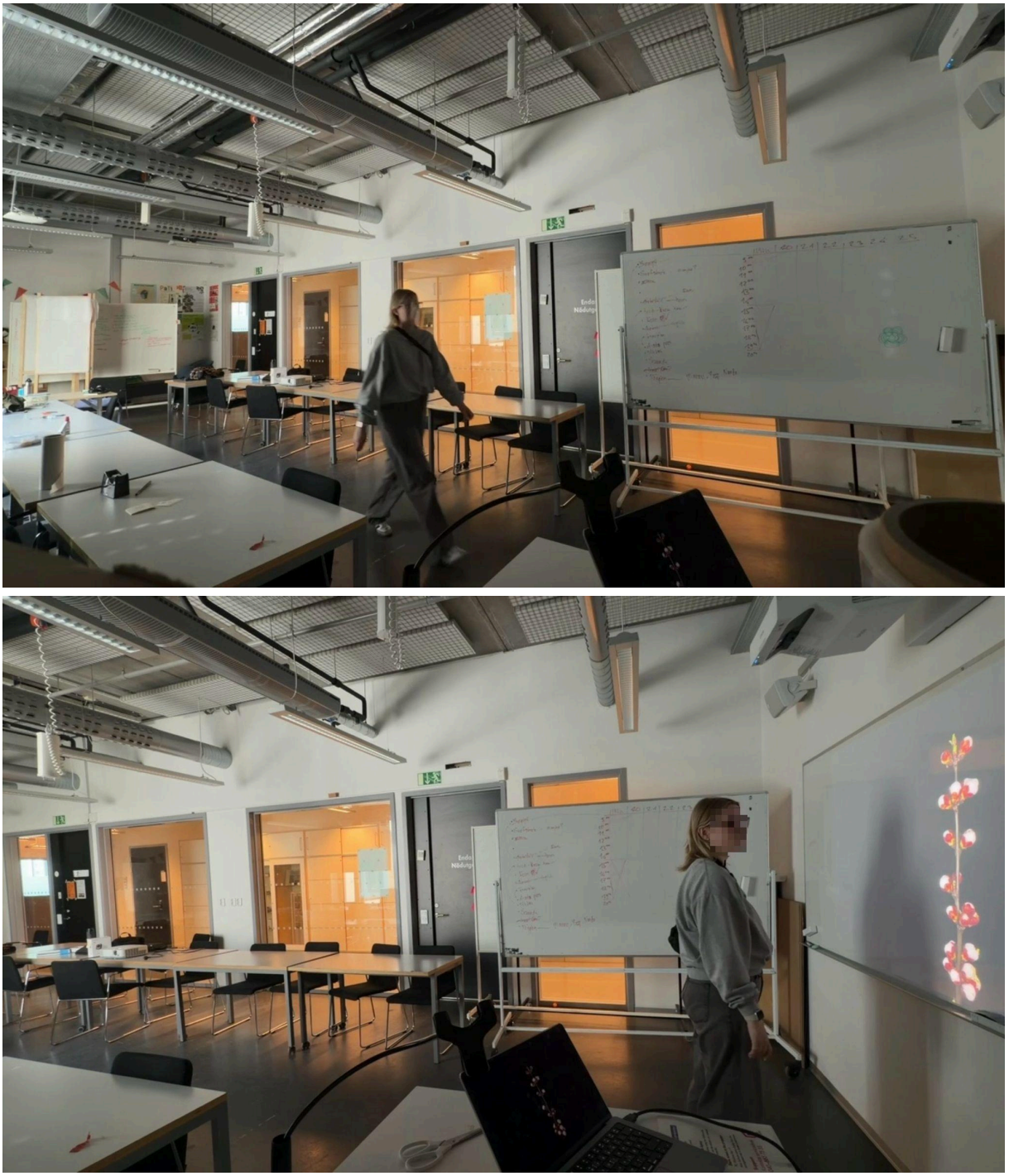


Figure 2 — *Preliminary Workshop Pathway in a Classroom*
(face blurred to ensure anonymity of participants)

4.4 Restructured Workshop

Following the preliminary Wizard of Oz workshop, the project was restructured around sound rather than visual projection. The flower video had begun to choreograph participant behaviour, so the revised installation removed visual feedback in order to allow participants to negotiate orientation more openly. This shifted the project toward a sound-based interaction in which non-normative movement, hesitation, and deviation could become the primary means of activation.

4.5 TouchDesigner

Intention and Premises

This stage explored how the installation could move from visual feedback toward a sound-based interaction. Following the preliminary Wizard of Oz workshop, projection was understood as too visually directive, because participants responded strongly to the flower imagery itself. The TouchDesigner sketch, therefore, tested whether bodily movement could instead generate an atmospheric audio response.

The interaction was carried out using TouchDesigner and Torin Blankensmith's MediaPipe TouchDesigner project, which provides MediaPipe-based tracking components for interactive audiovisual systems (Blankensmith, 2025). TouchDesigner was chosen because it supports real-time visual programming and rapid testing of interactive media systems.

Materials

- TouchDesigner
- MediaPipe TouchDesigner plugin
- Laptop
- Phone camera

Number of Participants

No participants were involved at this stage. This was a technical design phase conducted by the thesis author.

Location

Personal laptop / TouchDesigner workspace

Length

Several iterative testing sessions

4.5.1 Image Segmentation

Description

The first version used the MediaPipe image segmentation. After downloading the MediaPipe TouchDesigner project from GitHub (Blankensmith, 2025), the project first had to be extracted from the zipped folder so that the external files could load correctly. Once the project was working, the laptop webcam was connected through a Video Device in TOP.

The segmentation demo produced a colourful body mask from the live camera input. This visual output (Figure 3) was not used as a projection, but as data. A TOP to CHOP operator converted the segmentation image into numerical channel values, which were then processed through Analyse CHOP, Trail CHOP, and Math CHOP operators.

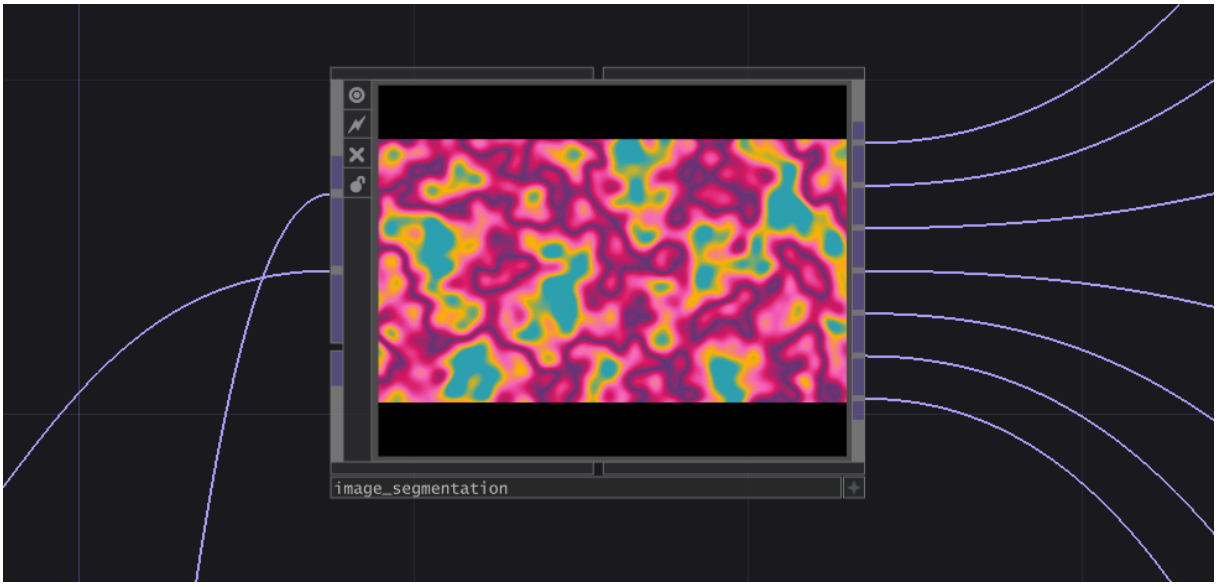


Figure 3 — Image Segmentation Output

The first version measured average changes in the segmentation image, but the results were too limited because the image values did not change meaningfully enough during bodily movement. A second version used a frame-difference method with Cache TOP, Cache Select TOP, and Difference TOP (Figure 4). This compares the current frame with a previous frame, allowing the system to detect visual change over time. The resulting values were then mapped to the volume of an audio file.

Takeaways

The image segmentation sketch showed that bodily movement could technically influence sound in TouchDesigner. However, it also revealed a major limitation: segmentation detected general motion intensity, but not orientation. It could not distinguish between straight movement, hesitation, oblique movement, lingering, or deviation from a hallway path.

This made the sketch useful as a test, but insufficient for the final installation. It clarified that the project needed a tracking system capable of analysing bodily position, trajectory, and orientation rather than simply detecting movement.

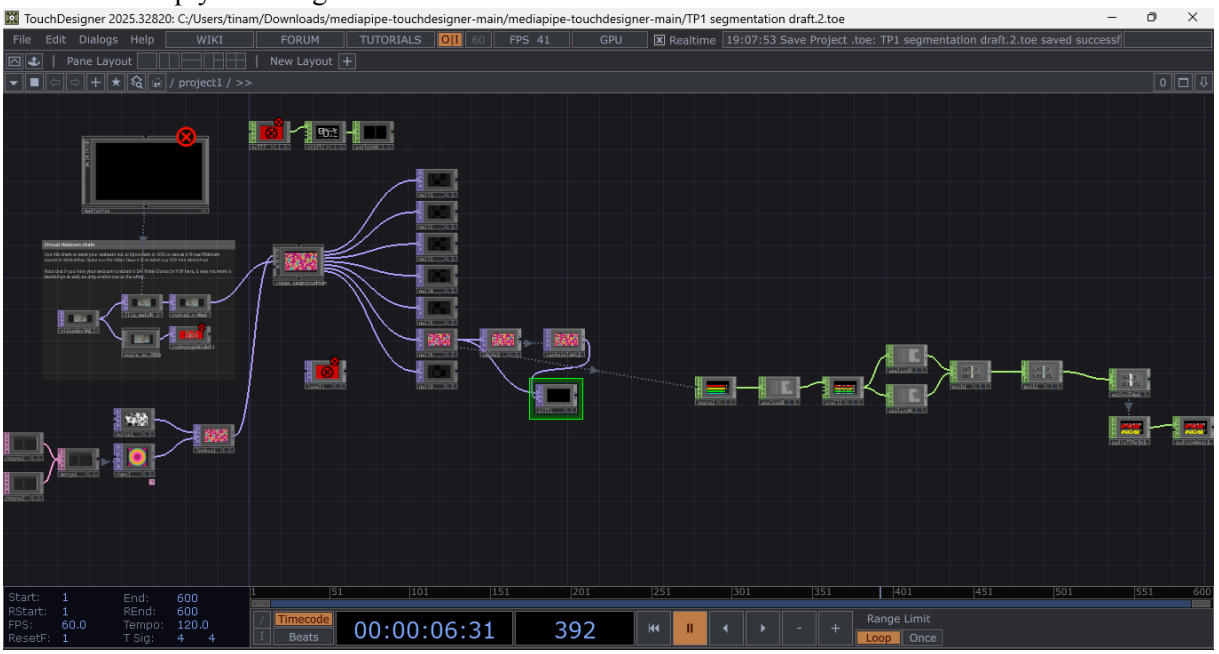


Figure 4 — *First Digital Sketch*

4.5.2 Pose, Head, and Hand Tracking

Description

The next digital sketch moved beyond image segmentation and toward tracking bodily orientation more directly (Figure 5). Using the MediaPipe TouchDesigner framework, the system was developed around three forms of bodily input: head orientation, walking trajectory, and hand or arm positioning.

A smartphone camera connected through DroidCam was used to capture the participant from a frontal hallway perspective. The tracking data was not used for visual output, but as behavioural input for a layered sound system.

The first layer focused on head orientation. Facial tracking was used to detect whether the participant was facing forward or redirecting their gaze sideways or upward. Looking away from the straight path slightly increased the sound.

The second layer focused on the walking trajectory. Pose-tracking data was intended to detect whether the participant moved in a straight line or deviated by slowing down, drifting sideways, curving, or interrupting their rhythm. This layer is directly connected to Ahmed's (2006) discussion of orientation and institutional straightening, where bodies are repeatedly directed along socially reinforced paths.

The third layer focused on hand and arm positioning. Hand tracking and pose estimation were used to identify moments where arms moved away from restrained or neutral bodily positions. Expanded arm movement contributed further to the sound intensity.

The audio logic was cumulative. One altered movement produced a small change, while multiple non-normative movements combined to create a louder and more intense soundscape. For example, redirected gaze, non-linear walking, and expanded arm movement together produced a stronger response than any single movement alone.

Takeaways

This digital sketch shifted the installation logic from detecting movement in general to detecting qualities of orientation. Rather than asking participants to "perform queerness," the system attempted to create conditions where non-normative bodily orientation became perceptible through interaction.

The sketch was not treated as a polished final technical system, but as an embodied digital sketch. Its purpose was to test how institutional movement patterns could be translated into interactive logic. Through this process, the project moved further away from visual interfaces and toward understanding interaction as bodily and spatial negotiation.

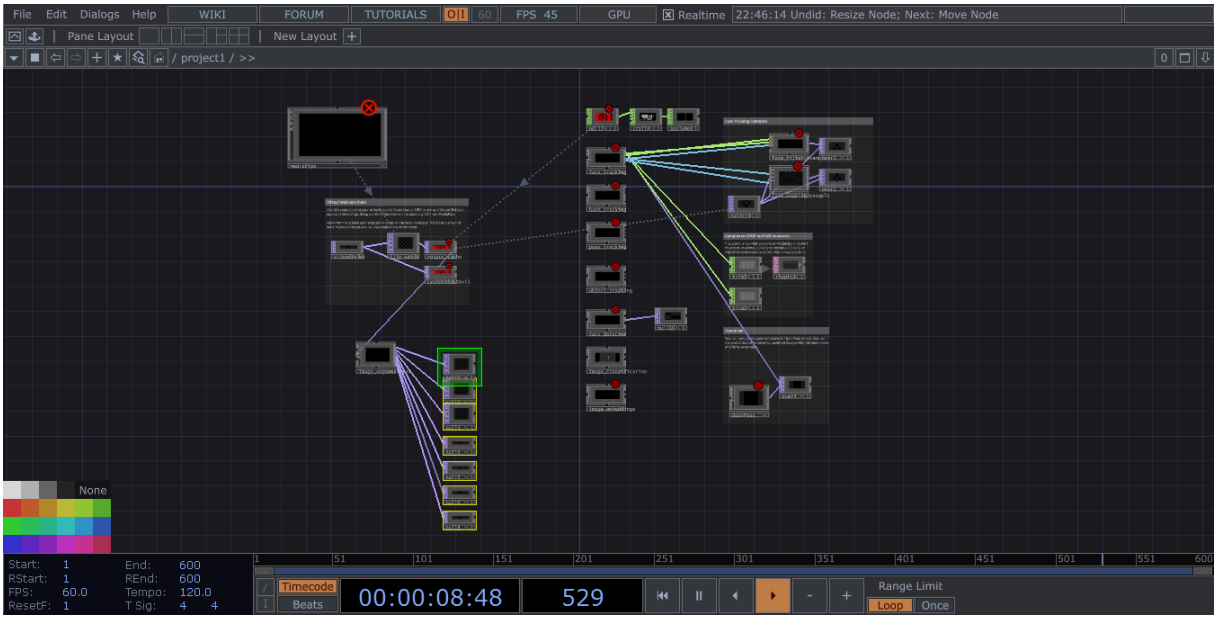


Figure 5 — *MediaPipe TouchDesigner example from Torin Blankensmith*

4.6 Final Workshops

4.6.1 Phase 1 — Analogue Trace and Oblique Movement

Intention and Premises

The first phase focuses on the straight lines of the institution (Ahmed, 2006). Using coloured tape on the floor, a path is established that mirrors the normative, efficient flow of the hallway. Participants are first asked to walk their habitual route. The premise of this phase is to prove that habitual behaviour in institutional spaces is the one of least resistance, most efficient and straightforward.

Materials

- Coloured tape
- Floor markings
- Printed prompts/instructions
- Notebook/documentation materials

Number of Participants

7

Location

Malmö University

Length

Approximately 2 minutes per participant

Description

Each participant was positioned behind a purple line taped onto the floor, and asked to get to the other one 10 meters straight ahead of them, down the hallway in the University.

Takeaways

All of the participants walked in a straight line from the first to the second tape. The premise of this phase was proved and further supported through the semi-structured interviews that followed the workshop. This acted as a base behaviour that the system was not responding to in the second phase.

4.6.2 Phase 2 — Digital Refusal and Sound Interaction

Intention and Premises

The second phase introduces a digital sound-based intervention responding to non-normative movement within institutional space. The phase focuses on how participants negotiate movement when interaction no longer relies on explicit visual cues or direct instruction. Rather than rewarding task completion, the installation responds subtly to bodily deviation

from habitual institutional movement patterns. The premise of this phase is for participants to achieve a sense of somatic consciousness and reflection, as well as use orientation as a design material.

Materials

- TouchDesigner
- Sound system
- Mobile phone
- Camera tracking system
- Computer

Number of Participants

3

Location

Malmö University

Length

5 minutes per participant

Description

Participants were positioned at the starting line again, and told to walk to the second tape – this time with the aim of activating the installation. They were not informed that it was sound-based, only that they should activate it through the task of going to the second taped line. (Figure 6)

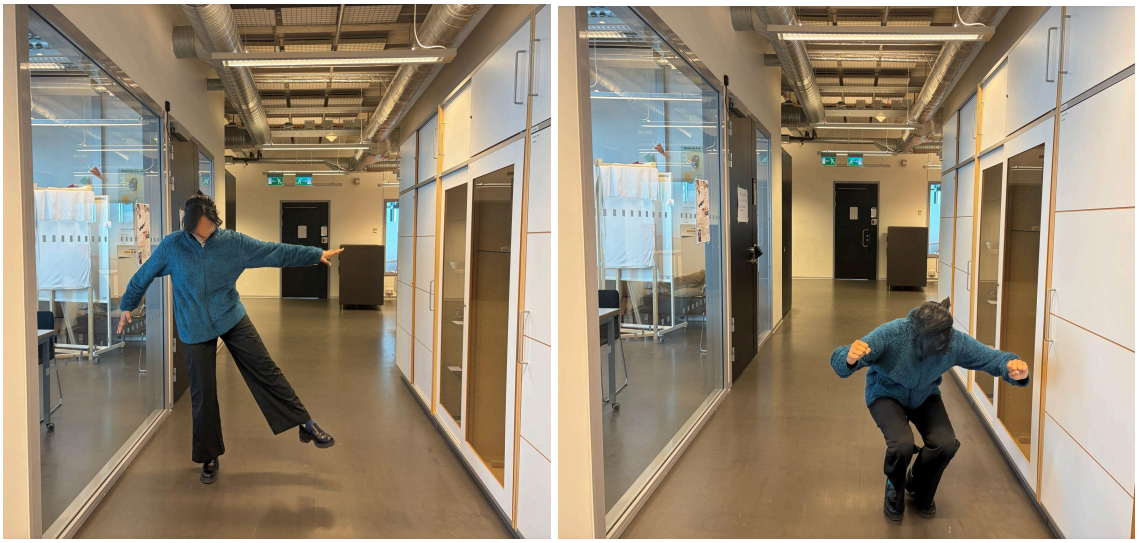


Figure 6 — *Participant 1 exploring non-normative movement*
(face blurred to ensure anonymity of participants)

Takeaways

Participants quickly realised that normative movement in a straight line, such as in the first phase, was not working now. They quickly turned to moving away from the straight line, jumping, going backwards, even squatting down or kicking the edges of the hallway. One of the participants did not even make it to the other tape, as they spent all of their time interacting with the sound around the middle point of the installation. The premise of this phase was looked at and discussed after having interviews with the participants.

4.6.3 Phase 3 — Reflective Drawing and Orientation Tracing

Intention and Premises

The final phase focuses on reflecting upon and externalising the experience of movement throughout the workshops (Sengers et al., 2005). Participants are invited to draw and trace their bodily orientations using layered drawings and spatial annotations based on their movement through the institutional environment.

Materials

- Tracing paper
- Printed site maps
- Pens and markers
- Transparent overlays

Number of Participants

3

Location

Malmö University

Length

5 minutes per participant

Description

Participants who partook in both the 1st and 2nd phase were now asked to draw with two colours. Firstly, the red colour represents their orientation and movement in the first phase. Secondly, the green marker represents their orientation – how they felt it somatically – during the second phase. (Figure 7)

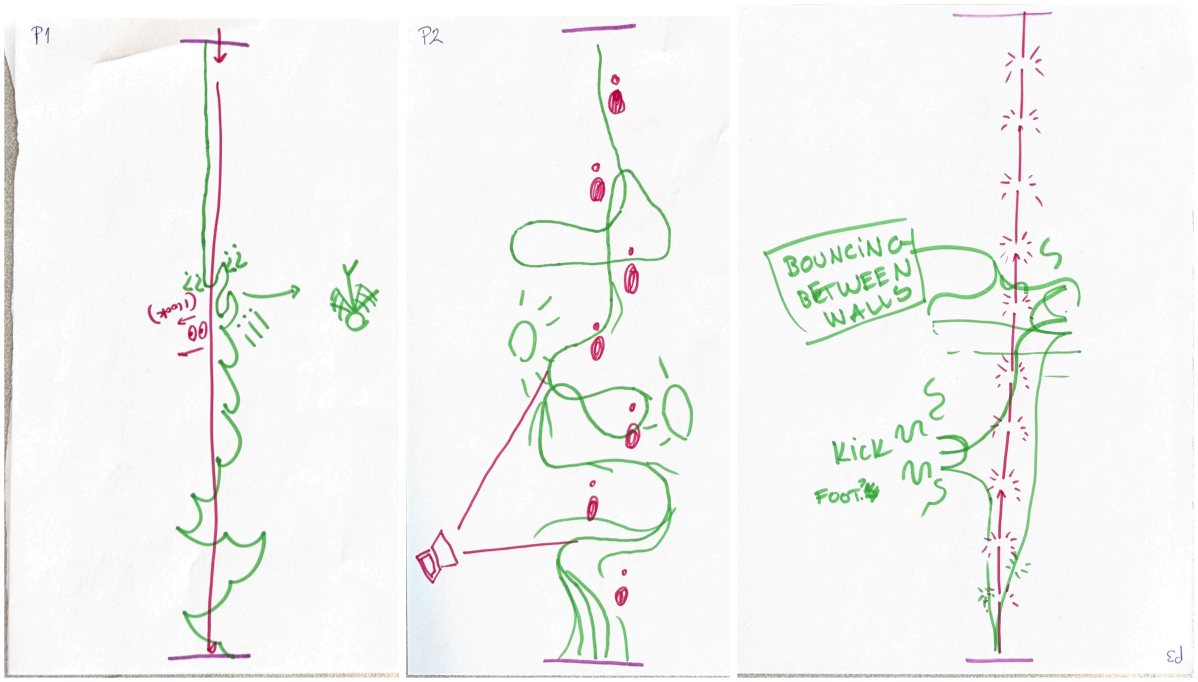


Figure 7 — *Drawings created by participants (chronological order 1-3, left to right)*

Takeaways

Orientation as observed is experienced within participants' bodies, and is felt as something that changes and is in relation to things outside of them. This provides orientation as a material that can be influenced, played with, or used in different ways.

4.7 Thematic Analysis of Semi-structured Interviews

Intention and Premises

Following the third workshop phase, where participants drew their movement trajectories and bodily orientations on paper, semi-structured interviews were conducted to further understand how participants experienced the installation. The drawing activity externalised movement visually, while the interviews allowed participants to reflect verbally on their somatic abilities, hesitation, visibility, social expectations, and movement within institutional space.

This phase connected participant-led reflection to process phenomenology, where lived bodily experience can be documented through drawing, notation, and reflective articulation (Kozel, 2015; Sengers et al., 2005). The interviews also followed a semi-structured format in order to keep the same core questions across participants while allowing space for individual reflection and unexpected themes to emerge (Kvale & Brinkmann, 2015).

Materials

- Participant movement drawings
- Interview question sheet
- Phone recorder
- Notebook/documentation materials
- Transcription notes

Number of Participants

3 participants

Location

Malmö University

Length

Approximately 10–15 minutes per participant

Description

After completing the drawing phase, participants were asked seven semi-structured interview questions. The interviews were transcribed and included in Appendix 2, and the questions are as written below.

1. Can you describe your experience of moving through the hallway during the installation?
2. At what point did you realise your movement was affecting the sound environment, and how did that influence the way you moved afterwards?
3. Did the installation make you more aware of your body, posture, or movement patterns? In what ways?
4. Were there any movements that felt natural, awkward, uncomfortable, or performative while interacting with the installation?
5. Did you feel encouraged to move differently than you normally would in a university hallway or institutional space?
6. Were there moments where you became aware of expectations regarding "proper" or socially acceptable behaviour while moving through the space?
7. Did the experience make you reflect differently on movement, visibility, or behaviour within institutional environments?

The questions were informed by the theoretical grounding of the thesis, particularly Ahmed's (2006) concept of orientation and institutional straightening. The aim was not to evaluate the installation as a usability system, but to understand how participants negotiated movement, visibility, and bodily expectation in relation to institutional space.

Takeaways

Several themes emerged across the interviews. First, all participants initially treated the hallway as a passage from one point to another, beginning with direct and efficient movement. This supported the project's premise that institutional spaces encourage habitual, straight, and goal-oriented movement.

Second, once participants realised that their movement affected the sound environment, they began experimenting with alternative orientations. Participants described moving sideways, zigzagging, jumping, walking backwards, flapping their arms, bouncing between walls, or creating wave-like movement patterns. These movements were not directly instructed, but emerged through participants' attempts to understand and negotiate the interaction.

Third, the interviews showed heightened somatic consciousness. Participant 1 became more aware of walking itself and described the hallway as usually functional. Participant 2 became especially aware of hand positioning and bodily orientation. Participant 3 focused more on feet, lateral movement, and the physical relationship to the walls. Across all three participants, movement that would normally remain automatic became conscious.

The interviews also showed different relationships to visibility and social normativity. Participants 1 and 2 described moments where unusual movement felt socially exposed or potentially inappropriate, especially when imagining being observed by others. Participant 3 offered a contrasting view, recognising that the movements were unusual but not experiencing them as uncomfortable or

socially problematic. This suggests that institutional expectations are not experienced equally by all participants.

Overall, the interviews demonstrated that the installation disrupted habitual hallway movement and encouraged participants to reflect on their bodily orientation. While the drawings showed where participants moved, the interviews revealed how those movements were experienced socially, affectively, and phenomenologically. Together, the drawings and interviews supported the thesis argument that orientation can be approached as a design material within Interaction Design.

4.8 Peer Reflections and Discussions

Intention and Premises

Alongside the workshops and the digital sketching process, peer discussion sessions were used as an additional reflective layer in the design process. The intention was to present the project to fellow Interaction Design students, discuss conceptual and practical challenges, and receive feedback on the direction of the work.

These sessions helped clarify the interaction logic of the installation and raised questions that had not yet been fully addressed, particularly around accessibility, bodily ability, ambiguity, and the risk of the installation privileging specific forms of movement.

Materials

- Project notes
- Workshop documentation
- Digital sketch description
- Peer feedback notes
- Visual documentation

Number of Participants

A small group of fellow Interaction Design students

Location

Malmö University/peer discussion setting

Length

Informal peer reflection sessions during the workshop and sketching period

Description

During these sessions, each student presented their project progress, current challenges, and unresolved questions. Feedback was then shared through discussion, written notes, and peer encouragement (Figure 8).

For this project, peers raised important questions regarding accessibility and bodily ability. Because the installation focused on movement and bodily orientation, some peers questioned whether it risked privileging able-bodied movement or more physically expressive forms of participation.

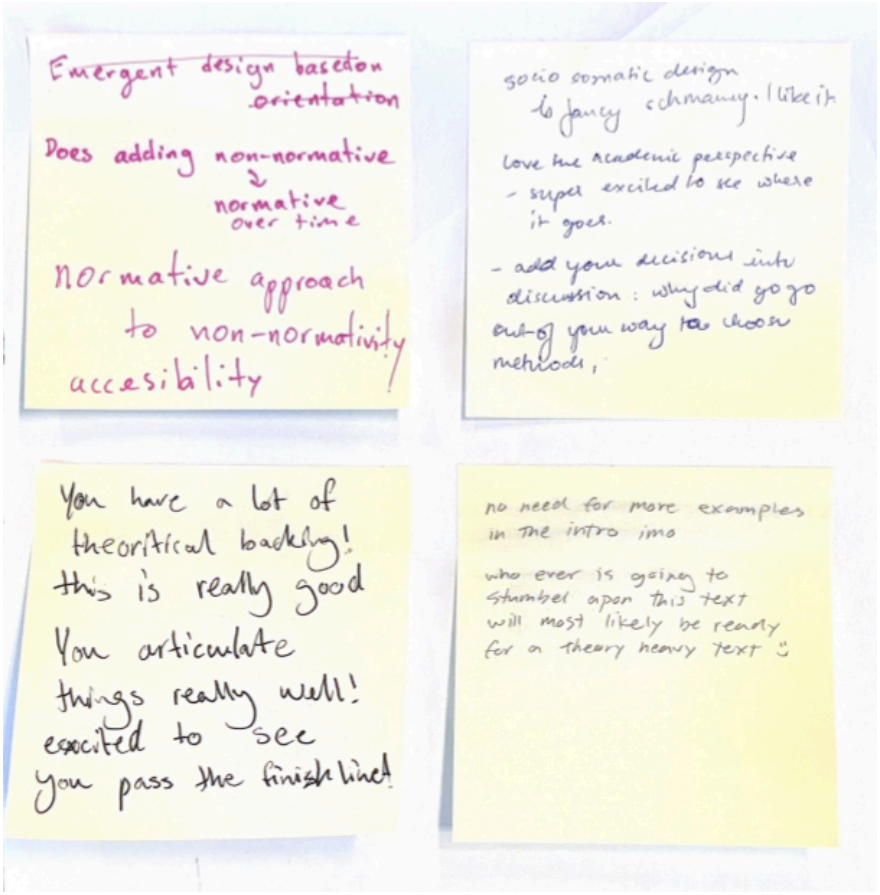


Figure 8 — *Notes from Interaction Design Master's colleagues*

Takeaways

The peer discussions helped clarify that orientation in this thesis is not understood as dependent on able-bodied movement, athleticism, speed, or physical capability. Instead, orientation is approached phenomenologically and relationally, following Ahmed's (2006) understanding of how bodies inhabit, approach, negotiate, and relate to space. From this perspective, orientation exists beyond any single bodily ability or physical manifestation.

The discussions also reinforced the importance of ambiguity within the installation. Peers responded positively to the absence of explicit instructions and to the subtlety of the sound-based interaction. This supported the decision to move away from visually directive feedback and toward a more atmospheric and exploratory interaction model.

Overall, the peer reflection sessions became a useful part of the iterative design process, helping refine the conceptual framing, accessibility considerations, and ethical positioning of the project.

5. DISCUSSION AND REFLECTION

5.1 Contributions

The final design outcome of this thesis is a sound-based embodied workshop installation situated within a university hallway. The installation does not respond to straight, efficient, and habitual movement, but instead becomes sonically activated through altered bodily orientations such as hesitation, drifting, slowing down, redirected attention, or non-linear

movement. Rather than functioning as a polished usability system, the outcome operates as a design intervention for investigating how orientation can be made perceptible through embodied interaction.

The purpose of the design outcome was not to produce a universal solution, but to explore the two research questions through situated design practice. The primary research question asked: How can interaction design facilitate a somatic consciousness of orientation in institutional spaces by refusing to respond to normative movement? Across the workshops, participants consistently began by moving through the hallway in direct, efficient, and goal-oriented ways. When the installation did not respond to this movement, participants began to hesitate, test, drift, jump, turn, or otherwise renegotiate their bodily orientation. In this sense, the refusal of normative movement created a temporary disruption that made participants more aware of their bodies, posture, visibility, and relationship to the institutional space.

The secondary research question asked: In what ways can orientation be a socio-somatic design material? Through the design process, orientation became something that could be explored, shaped, traced, amplified, and reflected upon. It appeared through participants' movement trajectories, paper drawings, sound interaction, hesitation, bodily self-awareness, and interview reflections. Orientation, therefore, emerged not only as a theoretical concept but as a material condition that designers can work with in space, through bodies, interaction, and response.

The contribution of this thesis can therefore be understood in two connected ways. First, it proposes orientation as a design material for Interaction Design. This means that designers may work with orientation not only as direction or navigation, but as a relational and embodied material shaped by movement, affect, visibility, social expectation, and spatial arrangement. In this project, orientation became available through walking, drawing, listening, pausing, deviating, and negotiating movement within the institutional spaces. This opens a design space where interaction designers can design with orientation rather than only designing for task completion, efficiency, or usability.

Second, the thesis offers a social contribution by exploring how interaction design may facilitate somatic consciousness of institutional orientations. The workshop installation invited participants to notice how their bodies move within university environments and how those movements are shaped by expectations of seriousness, efficiency, visibility, and appropriate behaviour. By encouraging non-normative movement through sound rather than direct instruction, the project created a temporary condition in which participants could consciously manipulate their own orientation and reflect on the institutional norms that usually remain unnoticed.

Together, these contributions show how Interaction Design can move between theoretical inquiry and embodied practice. The project does not claim to fully resolve institutional normativity or fill the gap between theory and practice, but it demonstrates how design can make such normativity felt, questioned, and temporarily reoriented. The final outcome is therefore important not as a contribution in itself, but as the design process through which these two contributions became possible: orientation as a design material, and somatic consciousness of institutional orientation as a social aim.

5.2 Limitations and Challenges

This thesis is limited by its small scope and situated context. The workshops involved a limited number of participants and took place within a specific university classroom and hallway. The findings should therefore not be generalised as universal claims about

movement, queerness, or institutional space. Instead, they should be understood as situated insights produced through a specific design process.

Another limitation concerns the author's interpretation. The project is shaped by my own understanding of normativity, orientation, and institutional behaviour. Although participant reflections supported several of the thesis arguments, the analysis remains interpretive and influenced by the theoretical framing of the project.

The project also contains a conceptual paradox. If non-normative movement is repeatedly encouraged, practised, or institutionalised, it may eventually become normative itself. This does not weaken the project, but it reveals an important tension: attempts to disrupt norms can also produce new norms over time.

Technical issues also shaped the development of the work. MediaPipe tracking, TouchDesigner integration, webcam quality, lighting conditions, and movement detection were not always stable. Early sketches detected general movement rather than orientation, which limited the precision of the interaction. These challenges influenced the final design and clarified the need for more robust tracking in future iterations.

Finally, the thesis prioritised phenomenological and experiential exploration over quantitative evaluation. It does not measure long-term behavioural change. Instead, its value lies in showing how interaction design can create temporary moments of somatic consciousness, reflection, and reorientation within institutional space.

6. CONCLUSION

This thesis explored whether orientation can be approached as a socio-somatic design material within Interaction Design, and how interactive workshops may facilitate somatic consciousness of institutional orientations. Rather than offering a definitive answer, the project should be understood as an initial exploration into how bodies, spaces, technologies, and social expectations interact within university environments.

Drawing from queer phenomenology, embodied interaction, ambiguity, somaesthetic interaction design, and Research through Design, the thesis investigated how institutional spaces encourage certain habitual movements while making others feel unusual, visible, or inappropriate. Through informal conversations, workshops, reflective drawings, TouchDesigner, sound-based interaction, and semi-structured interviews, the project examined how participants experienced and negotiated movement within institutional hallways.

The final workshop installation did not aim to produce a polished or universal system. Instead, it functioned as a situated design intervention that refused to respond to straight, efficient, and normative hallway movement. This refusal created moments of hesitation, experimentation, and bodily awareness. Participants began to test alternative movements, reflect on their visibility, and notice how institutional spaces quietly shape what feels proper, efficient, or acceptable.

In this sense, the thesis suggests that orientation may be considered as a design material. Orientation appeared not only as direction or navigation, but as something relational, affective, bodily, and socially shaped. It became visible through walking, pausing, drifting, drawing, listening, and reflecting. However, this proposition remains open and unfinished. The thesis only begins to investigate what designing with orientation might mean, and further work would be needed to explore how this material could be used across different spaces, bodies, technologies, and social contexts.

The project also suggests a social contribution. By encouraging participants to consciously manipulate their movement, the workshop opened a small space for questioning how bodies are expected to behave in university environments. This does not resolve institutional normativity, nor does it provide a stable method for producing non-normative

space. Rather, it shows how design can make such norms momentarily felt, questioned, and negotiated.

Ultimately, this thesis should be understood as a first step into expanding Queer HCI space. It scratches the surface of how orientation, normativity, somatic awareness, and institutional space might be explored through interaction design. Its value lies not in finality, but in opening a direction: toward interaction design practices that not only optimise movement through space, but also allow bodies to notice, question, and reorient themselves within it.

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A special thank you to queer people today and before, whose orientation in every meaning of that word shapes the lived experience.

Thank you.

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10. APPENDIX

1 — Notes from the preliminary Wizard of Oz workshop

Workshop Overview

The preliminary workshop was conducted on the 22nd of April as an early embodied probe. The workshop took place inside a classroom at Malmö University and involved three participants interacting individually with the installation setup.

Installation Setup

The installation consisted of:

- a classroom space,
- a projector,
- a projected video of a blooming flower displayed on the classroom whiteboard,
- and manually controlled interaction behaviour.

Participant Interactions

Participant 1

Participant 1 initially approached the interaction from a highly functional and efficiency-oriented perspective. After receiving the instructions, the participant immediately attempted to walk directly toward the whiteboard as quickly as possible.

When the installation failed to react, the participant expressed frustration and confusion regarding the lack of feedback from the system. During the interview, the participant described the interaction as “useless” when no response occurred and explained that they expected a direct relationship between task completion and system activation.

After several failed attempts, the participant gradually began reconsidering the interaction logic and reflected that “maybe a straight line is not triggering.” This led them to experiment with alternative bodily movement patterns.

The participant eventually approached the whiteboard using zig-zag movement combined with hopping motions. Once the installation activated, the participant immediately associated activation with non-straight movement, explaining:

> “If I move straight — nothing, if I do something else — it reacts.”

During the interview, the participant reflected upon how the interaction disrupted their assumptions regarding movement within institutional space, stating:

> “The default is to always go from point A to point B.”

The participant described straight movement as automatic, habitual, and efficiency-driven.

Participant 2

Participant 2 similarly began the interaction by walking directly toward the whiteboard "without any thinking" and described their initial behaviour as simply moving "straight to the goal."

However, unlike Participant 1, Participant 2 became increasingly aware of being observed during the interaction. As the installation repeatedly failed to react, the participant described growing feelings of uncertainty, bodily tension, embarrassment, and self-consciousness.

After several failed attempts, the participant gradually began experimenting with more performative and exaggerated bodily movements. These included:

- turning in circles,
- spinning,
- and performing pirouettes inspired by the projected flower imagery.

The participant later reflected that these movements felt "weird" and inappropriate within a classroom environment because they attracted attention and disrupted expected classroom behaviour.

During the interview, the participant explained:

> "I wouldn't move like this if the classroom was full of people."

The participant repeatedly associated classroom environments with seriousness, restraint, and appropriate bodily conduct. They explained that performing pirouettes or unusual movement patterns felt embarrassing because such behaviour conflicted with how one is socially expected to move within institutional space.

The participant also reflected that their movement became increasingly performative once they realised the installation was refusing normative movement patterns.

Participant 3

Participant 3, who identified as queer, demonstrated a different relationship toward bodily orientation and visibility from the very beginning of the workshop.

Even during the initial observation phase, the participant avoided walking directly through the centre of the classroom. Instead, they instinctively moved toward the edge of the room while approaching the whiteboard.

During the interview, the participant explained:

> "If I walk to the centre, I feel like I'm gathering attention, and I don't want that."

Unlike the other participants, whose movement initially followed straightforward linear paths, Participant 3 already demonstrated heightened awareness of:

- visibility,
- observation,
- bodily positioning,
- and spatial attention.

The participant repeatedly reflected upon feeling conscious of being observed throughout the interaction. They explained that they were not only thinking about how to complete the task, but also whether they were "doing what the observer wants."

The participant described becoming overly aware of:

- their walking pace,
- body positioning,
- and movement trajectory.

Walking itself gradually stopped feeling automatic and instead became consciously negotiated and performed.

One particularly notable observation was the participant's instinctive avoidance of occupying the centre line directly in front of the projected image. The participant explained that standing directly in front of the projection felt like "taking the attention away."

Instead, they diverted toward the edges of the room while approaching the whiteboard. The participant later connected this behaviour to broader spatial habits, explaining that they often avoid central positions within spaces such as:

- museums,
- presentations,
- classrooms,
- and gallery environments.

They described frequently positioning themselves toward the edges of paintings or rooms in order to minimise visibility and avoid becoming the centre of attention.

Semi-Structured Interview Themes

Following the interaction, semi-structured interviews were conducted focusing on:

- bodily awareness,
- movement,
- visibility,
- institutional expectations,
- discomfort,
- uncertainty,
- and orientation.

The interviews gradually shifted away from usability-oriented questioning and instead focused more closely on somatic and affective experience.

Participants were asked questions regarding:

- how their body prepared to move,
- how they interpreted the installation,
- what they felt when the installation did not react,
- what emotional or bodily changes occurred before altering movement,
- and how they perceived themselves within classroom space during the interaction.

Key Insights and Reflections

Several important insights emerged from the workshop.

Across all participants, straight movement initially emerged as automatic, habitual, and unquestioned. Participants consistently described their first instinct as moving efficiently from “point A to point B.”

The workshop also revealed strong connections between:

- movement,
- visibility,
- social expectation,
- and institutional normativity.

Participants repeatedly described feeling that classrooms demand seriousness, restraint, and normative bodily behaviour. Non-normative movement patterns such as spinning, zig-zagging, or pirouettes were frequently associated with embarrassment, attention, or social discomfort.

The queer participant demonstrated significantly heightened awareness of bodily visibility and spatial positioning even before the interaction began disrupting normative movement patterns. This contrasted with the non-queer participants, whose awareness of these structures primarily emerged only after the installation repeatedly refused their habitual movement.

These observations suggested that orientation within institutional environments is not solely spatial, but deeply social, affective, and relational.

The workshop further revealed limitations within the original installation concept. Simply asking participants to “walk non-straight” risked reducing orientation into a singular performative action rather than investigating the broader relationship between bodies, institutions, visibility, and normative spatial behaviour.

Following this workshop, the project gradually shifted away from a singular classroom installation toward a wider investigation of institutional orientation throughout university environments.

The workshop therefore became an important turning point in the development of the thesis, directly informing the restructuring and expansion of the later workshop phases.

2 — Post-workshop semi-structured interviews transcript

Participant 1

1.Can you describe your experience of moving through the hallway during the installation?

At first, I was walking mostly straight through the hallway. At one point, I passed by a window and looked inside the classroom. Then I started moving more slowly because I was trying to understand how the installation worked. I noticed the sound changed sometimes, but not consistently, so I became confused about what was causing it. I first thought the installation was position-based, like I had to stand in a certain spot for the sound to activate. I started moving backwards and forwards to test this. Then I began reaching out with my hands to see whether movement itself was affecting the sound. Eventually, I realised that when I moved differently, especially when I started flapping my arms or jumping, the sound became louder. After that, I began hopping and zigzagging through the hallway while continuing to flap my hands. The experience became much more playful after I understood that unusual movement was activating the installation.

2. At what point did you realise your movement was affecting the sound environment, and how did that influence the way you moved afterwards?

At first, I stopped in place and then tried moving backwards slightly because I thought the installation responded to position. But when I stopped moving, there was no sound. Then I moved forwards and backwards again and noticed changes in the audio. That was when I started thinking, “What happens if I do something different?” Usually, when I walk through a hallway, I just walk straight toward wherever I am going. The only things I might normally do are let someone pass me or briefly look into a classroom. Otherwise, I move directly through the space. Once I realised the installation reacted to unusual movement, I started flapping my hands, jumping, and generally trying to be as loud and exaggerated as possible with my body. I intentionally moved differently from the way I would normally move in a hallway.

3. Did the installation make you more aware of your body, posture, or movement patterns? In what ways?

Yes. It made me realise that I normally do not think about hallways as spaces where I could jump, hop, or play. During the installation, I became very aware of how I was moving because I was trying to activate the sound. At the beginning, I moved cautiously and consciously because I knew there was something in the hallway that I was supposed to interact with, even though I did not initially understand how it worked. I became aware of the difference between my normal way of walking and the movements I was experimenting with during the interaction. Afterwards, I actually felt excited by the experience. It made me feel like I could move differently in these spaces.

4. Were there any movements that felt natural, awkward, uncomfortable, or performative while interacting with the installation?

At first, my movements felt cautious and very self-aware. I was consciously thinking about how I was walking because I was trying to trigger the installation somehow. However, once I understood that the installation responded to unusual movement, it stopped feeling uncomfortable. Instead, it became fun and playful. I would describe the movements as performative in the sense that I was intentionally experimenting with my body and movement in the space, but not in a negative way. The experience made me realise the difference between how I normally move through a hallway and how I could move if I allowed myself to act differently.

5. Did you feel encouraged to move differently than you normally would in a university hallway or institutional space?

Yes, definitely. Normally, I would never skip, hop, or jump through a hallway. But during the installation, I felt encouraged to do exactly that. Now I almost feel like I want to skip to class sometimes. The installation made me feel like movement in these spaces does not always have to be restrained or serious.

6. Were there moments where you became aware of expectations regarding “proper” or socially acceptable behaviour while moving through the space?

Yes. At one point, an older person walked past while I was interacting with the installation. When they approached, I immediately became aware that my behaviour might not be considered socially acceptable in a hallway. I stopped what I was doing for a moment and moved aside to let them pass. If I had just been walking normally, I probably would not have reacted so strongly. But because I was already moving unusually, I suddenly became very aware of how visible I was. After they passed by, I continued jumping and moving anyway. At one point they even turned around to look back at me. At first the movements felt unnatural

because they did not match socially acceptable hallway behaviour, but the longer I continued, the more normal they began to feel to me.

7. Did the experience make you reflect differently on movement, visibility, or behaviour within institutional environments?

Yes. For me, the experience connected to learning how people behave within institutional and public spaces. When I moved to Sweden, I had to learn unspoken social rules, such as which side people walk on, how people position themselves in elevators, and how people generally behave in shared spaces. Over time, I became used to those behaviours and expectations. But the installation made me realise that these rules can also be broken without anything bad happening. It made me feel like I could behave differently within institutional spaces if I wanted to. For example, if I wanted to sit in an empty classroom, lie down on a sofa in a public area, or simply move differently through the hallway, that would actually be okay. The installation made me reflect on how strongly these social expectations shape movement, even when they are never explicitly stated.

Participant 2

1. Can you describe your experience of moving through the hallway during the installation?

At first, I still understood my goal as getting from one side of the hallway to the other, so I naturally began by moving forward in that direction. However, because I expected something to happen, I also started experimenting with movement. I tried moving sideways and thought about touching the wall, although I assumed touching it probably would not actually affect the installation. I also wondered whether there was a difference between walking towards another person or away from them (the installation conductor), since there was a social aspect to the interaction as well. Eventually, I started thinking about movement in relation to waves or the ocean. I noticed that moving sideways or in non-linear ways seemed to amplify the sound, while continuing in a straight forward direction made the sound calmer again. This led me to experiment with wave-like movement patterns while still progressing through the hallway. The interaction became playful because it felt like my movement was shaping the sound environment in real time.

2. At what point did you realise your movement was affecting the sound environment, and how did that influence the way you moved afterwards?

I realised it quite early, especially when I first moved sideways away from my normal path. I was already expecting the installation to respond somehow, but once I changed direction and then turned again, I started understanding the relationship between movement and sound. That was when the “ocean” metaphor came into my mind. After that, I intentionally tried to move in ways that resembled waves or flowing motion because I liked the way the sound responded to those movements. The installation influenced me to experiment more consciously with my orientation and movement patterns instead of simply walking straight through the hallway.

3. Did the installation make you more aware of your body, posture, or movement patterns? In what ways?

Yes. I became especially aware of what I was doing with my hands. At first, I was unsure whether my hands affected the sound or not, so I consciously experimented with them. Normally, if I were simply walking through the hallway, I would move in a straight line and continue directly toward my destination. I would not normally turn around or walk backwards. During the installation, however, I became much more conscious of how I was positioning and moving my body. I also noticed that I became less focused on the surroundings, such as

the classrooms I was passing, because my attention shifted toward the sound and my own bodily orientation. During the second interaction, I also became more aware of the presence of the researcher in the space, since they were connected to the sound experience itself.

4. Were there any movements that felt natural, awkward, uncomfortable, or performative while interacting with the installation?

The most awkward aspect for me was not knowing what to do with my hands. I became very conscious of them because I was unsure whether they influenced the sound. At the same time, there was a tension between wanting to move directly from point A to point B and wanting to experiment with the installation. It felt slightly uneasy because I had to improvise and intentionally think of movements that were different from my normal behaviour in order to make something happen. The interaction therefore felt performative in the sense that I was actively experimenting with movement that I would not usually perform in a hallway.

5. Did you feel encouraged to move differently than you normally would in a university hallway or institutional space?

Yes. The installation encouraged me to interrupt my normal movement patterns and experiment with movement in ways I would not normally do. For example, I began walking sideways, turning around, and changing orientation within the hallway. These are movements I would not normally perform in an institutional corridor because the usual expectation is simply to move efficiently toward your destination. The installation created a reason to explore movement differently.

6. Were there moments where you became aware of expectations regarding "proper" or socially acceptable behaviour while moving through the space?

Yes. I became very aware that some of the movements felt socially unusual within a hallway environment. For example, touching the wall or suddenly making a U-turn in the middle of the hallway to intentionally increase the sound felt strange. I imagined that if someone observed me without hearing the audio themselves, my behaviour would probably appear confusing or irrational. That awareness made me realise how strongly hallways are associated with socially regulated movement and expected behaviour.

7. Did the experience make you reflect differently on movement, visibility, or behaviour within institutional environments?

Yes. The experience made me reflect on what I normally pay attention to when moving through institutional spaces. Usually, when I walk through a hallway, I pay attention to what is happening around me, such as looking into classrooms or observing other people. During the installation, however, my attention shifted inward toward my own body and orientation in space. Instead of focusing on the surrounding environment, I became much more focused on my personal movement and how my body related to the sound interaction. The installation therefore changed my awareness of both movement and spatial behaviour within the institutional environment.

Participant 3

1. Can you describe your experience of moving through the hallway during the installation?

During the first walk, I immediately became very self-conscious because I knew I was being observed. That made me much more aware of my walking itself, especially the sound my shoes made on the floor. My shoes that I am wearing today are naturally loud, but normally that sound fades into the background. During the installation, however, it became very

noticeable and amplified in my awareness. During the second walk (the Touch Designer part), I noticed that I gravitated more toward one side of the hallway, particularly toward the side where you were standing (the workshop conductor). At first, I walked straight ahead and did not notice major changes in the sound. Then, when I moved sideways toward the wall, it felt like the sound became louder. Because of that, I started experimenting. I tested whether kicking the wall or moving close to it would influence the sound. Kicking the wall seemed to make the sound louder, so I continued experimenting by bouncing between the walls and moving side to side rather than simply forward. At some point, I realised that moving closer forward did not seem to change much, whereas lateral movement created more noticeable reactions. That is why I continued moving from side to side instead of walking directly toward the end of the hallway.

2. At what point did you realise your movement was affecting the sound environment, and how did that influence the way you moved afterwards?

I realised fairly early, around one or two metres into the hallway, that my movement was connected to the sound environment. Specifically, I realised that the interaction seemed more dependent on the movement of my feet and my side-to-side movement rather than simply moving forward or backward. Once I understood that, I intentionally started experimenting more with lateral movement and kicking motions to see how the installation would respond. I didn't even consider the task of walking to the other purple tape on the floor.

3. Did the installation make you more aware of your body, posture, or movement patterns? In what ways?

It did not necessarily make me more aware of my posture itself, but it made me very aware of my feet and the way I was moving them. For example, I became conscious of how I was rotating my body toward the walls in order to kick or interact with them. So the installation increased my awareness of bodily orientation and movement patterns within the hallway space.

4. Were there any movements that felt natural, awkward, uncomfortable, or performative while interacting with the installation?

The kicking definitely felt unusual because it is not something you normally do while walking through a hallway. At the same time, moving side to side and bouncing between the walls felt somewhat exploratory, almost like a new student investigating the hallway and looking at displays or objects around them. However, repeatedly moving between both walls felt more performative because it was clearly connected to activating or testing the installation itself rather than simply exploring the space naturally.

5. Did you feel encouraged to move differently than you normally would in a university hallway or institutional space?

Yes. The installation encouraged me to experiment with movement that I would not normally perform in a hallway environment, especially moving laterally and interacting physically with the walls. Instead of simply walking directly from one point to another, I became more focused on exploring the space through movement and testing how the installation reacted.

6. Were there moments where you became aware of expectations regarding "proper" or socially acceptable behaviour while moving through the space?

Personally, I do not strongly associate walking with socially acceptable or unacceptable behaviour, aside from basic expectations such as not bumping into people. Therefore, while some of the movements, such as kicking the walls, might conventionally appear unusual, I

did not personally experience them as socially inappropriate. I was mainly performing those actions because the interaction was enjoyable and playful. I also felt that if the installation became permanent, these movements could eventually become normalised as part of that hallway environment.

7. Did the experience make you reflect differently on movement, visibility, or behaviour within institutional environments?

The experience made me think about how visible unusual movement might appear to others within institutional spaces. I recognised that other people would probably notice me more if I moved in those ways within the hallway. However, I did not attach any negative emotion or discomfort to that increased visibility. From my perspective, especially as a designer, standing out or behaving differently within space already feels relatively natural.